SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	V - Photogrammetry Surveying

I. Introduction to Photogrammetry

Photogrammetry is the science and art of obtaining ***reliable measurements and maps from photographs*** . The word comes from photo (light), gramma (record) and metron (measure).

- Photographs are taken from the air (aerial) or from the ground (terrestrial); measurements on the photographs give the positions, shapes, sizes and heights of objects.
- Used for topographic mapping, contour maps, DEMs, road / railway / canal alignment, land use, forestry, geology, urban planning, military intelligence and archaeology.

Advantages

- Covers large areas quickly; the permanent record of the ground at an instant of time.
- Access to inaccessible and dangerous terrain; less fieldwork; economical for large projects.
- Data can be processed repeatedly; both planimetric and height data from the same photographs (stereo pairs).

Limitations

- Weather and light dependent; heavy initial equipment cost; needs ground control points; poor under dense forest and cloud.

2. Basic Concepts and Classification

Basis	Classification
Position of the camera	**Aerial** (from aircraft, drone, satellite) and **Terrestrial** (from ground stations)
Camera axis (aerial)	**Vertical** (tilt not more than 3 deg), **Tilted / near-vertical** (tilt 3 - 30 deg), **Oblique** (axis intentionally tilted: low oblique does not show the horizon; high oblique shows the horizon)
Equipment used	Analogue (film), analytical, **digital photogrammetry**
Purpose	Metric (measurement) and interpretative photogrammetry
Type of film	Panchromatic, infrared, colour

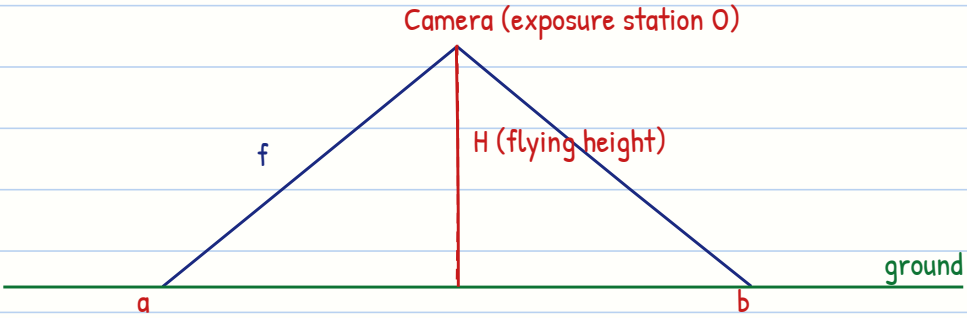


Fig 1: vertical aerial photograph geometry

Terms used on an aerial photograph

Term	Meaning
Principal point (p)	Foot of the perpendicular from the lens centre (rear nodal point) on the photo plane - the centre of the photograph
Nadir point (n)	Point where the plumb line through the exposure station meets the photograph
Isocentre (i)	Point on the photograph that lies on the bisector of the tilt angle
Exposure station (O)	Position of the camera lens at the time of exposure
Fiducial marks	Marks at the sides / corners of a photograph; their intersection = the principal point; used for measurements

Focal length (f)	Distance from the lens to the photographic plane (typically 152 mm, 210 mm, 88 mm)
Flying height (H)	Height of the exposure station above the datum (mean ground level)
Tilt (t)	Angle between the camera axis and the vertical
Parallax	Apparent displacement of an object due to change in the observer's position

3. Perspective Geometry of an Aerial Photograph

An aerial photograph is a ***central (perspective) projection*** : all the rays pass through the lens. A vertical photograph of a flat terrain is a scale reproduction of the ground.

Scale of a vertical photograph

Scale S = f / (H - h) = photo distance / ground distance
For flat ground (h = 0): S = f / H
Representative fraction RF = 1 : (H - h) / f

- h = elevation of the ground point above the datum; so the scale varies with the height of the terrain (higher ground -> larger scale, because it is nearer to the camera).
- ***Average scale*** uses the average terrain elevation h_avg: S_avg = f / (H - h_avg).
- If a map or a known ground distance is available: S = (photo distance ab) / (ground distance AB).

Ground length = photo length / S ; Area on ground = photo area / S^2

Example

A vertical photo is taken with f = 152 mm from H = 3200 m above datum. A point at h = 400 m elevation. Scale = 0.152 / (3200 - 400) = 1 / 18,421. An 8 cm line on the photograph = 8 x 18,421 / 100 = 1473.7 m on the ground.

Coordinates of the points on a vertical photo

- Photo coordinates (x, y) measured with the principal point as origin; the ground coordinates are:

$$X = x (H - h) / f \; ; \quad Y = y (H - h) / f$$

Ground length AB = sqrt[(XB - XA)^2 + (YB - YA)^2]

4. Relief Displacement

On a vertical photograph, the images of points above or below the datum are ***displaced radially*** from their true plan positions. This is ***relief displacement*** . Tall objects (towers, buildings, trees) appear to lean outwards from the principal point (the nadir).

$$d = r h / H = r' h / (H - h)$$

d = relief displacement; r = radial distance of the top of the object from the principal point;

h = height of the object; H = flying height above the base of the object

- Displacement is ***outward*** for a point above the datum and ***inward*** for a point below the datum.
- d increases with r (distance from the centre) and with h; d is ***zero at the principal point*** (r = 0) and zero for h = 0.
- Height of an object from the relief displacement: ***h = d H / r*** (the base and the top radial distances from the principal point are measured on the photograph).

USE:

Relief displacement can be used to find heights of tall objects (chimneys, towers); and it is also the basis of stereoscopic viewing and the height measurement. It decreases with higher flying height.

Example

A chimney top is 90 mm from the principal point on the photograph, and the relief displacement = 3.6 mm. Flying height above the base = 1800 m. Find the height of the chimney.

$$h = d H / r = 3.6 \times 1800 / 90 = 72 \text{ m}$$

5. Tilt Displacement

A ***tilted photograph*** has its optical axis inclined to the vertical by a tilt angle t. The photograph is no longer a scale model of the ground: the scale varies from side to side and the points are displaced ***radially from the isocentre*** .

- Tilt displacement is measured radially from the isocentre; it grows with the square of the radial distance and with the angle of tilt (approximately d_t = r^2 sin t cos phi / f for small tilts).
- Directions: points on the upward side of the isometric parallel are displaced ***away from*** the isocentre, on the lower side ***towards*** the isocentre; points on the isometric parallel (the line through the isocentre perpendicular to the principal line) are not displaced.

Tilt displacement (approx.): d_t = (r^2 / f) sin(t) cos(phi)
r = radial distance from the isocentre; phi = angle measured from the principal line

- Principal plane: vertical plane through the camera axis; principal line: its trace on the photograph; isometric parallel, axis of tilt, true horizon line - the three key lines of a tilted photograph.
- Tilt is removed by ***rectification*** (producing an equivalent vertical photo).

6. Terrestrial Photogrammetry

The photographs are taken from ***fixed stations on the ground*** with a ***phototheodolite*** (a camera mounted on a theodolite) or a metric camera. Used for close-range work: buildings, bridges, dams, mines, landslides, archaeology and accident sites.

- ***Phototheodolite:*** a camera + theodolite - gives the direction of the camera axis and the horizontal angle.
- ***Stereometric camera:*** two cameras fixed on a base bar of a known length (e.g. 1.2 m) - to give a stereo pair of the same area.
- Camera axes: (1) ***Horizontal / level*** and normal to the base line (***normal case***); (2) ***Parallel*** , (3) ***Convergent*** , (4) ***Divergent*** - the axes may be parallel or oriented at an angle.

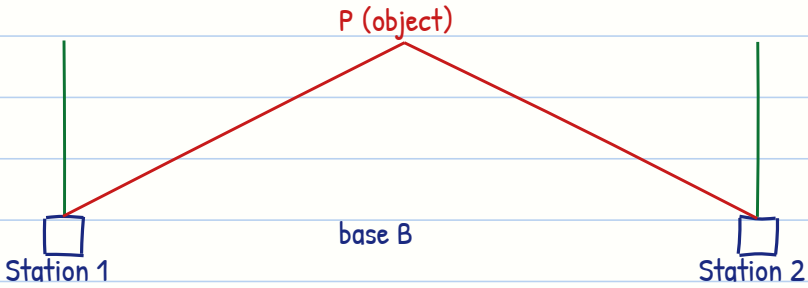


Fig 2: terrestrial stereo photographs (convergent pair)

Distance (Y) = (B x f) / (x_left - x_right) [normal case]

B = base length; f = focal length; the difference of x-coordinates = parallax p

- Distance from the camera: $Y = B f / p$; Height: $Z = Y y / f$; Lateral coordinate: $X = Y x / f$.
- Advantages: low cost, quick, exact records of the object for measurement later, and no contact with the object. Limitations: obstruction in view, shadow, hidden areas.

7. Flight Planning for Aerial Photography

A good flight plan is essential for efficient and economical aerial mapping. The ***flight map*** is prepared on an existing map showing the flight lines, the exposure stations and the scale.

Factors considered

- Purpose and the scale of the final map; required accuracy; area, shape, terrain relief; camera focal length and the format (e.g. 23 x 23 cm); weather and season (sun angle 30-40 deg, no clouds, snow, leaves).
- Flight height, speed of the aircraft, overlap, number of strips and the number of photographs.

Overlaps

- ***Forward (longitudinal) overlap:*** 60 % (55-65 %) between the successive photographs in a strip - required for stereo vision.
- ***Side overlap (lateral):*** 20 - 30 % (typically 25 - 30 %) between the adjacent strips - avoids gaps.

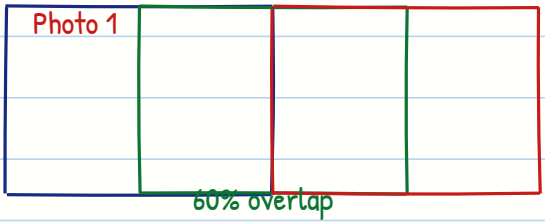


Fig 3: forward overlap of photographs

Design formulae

Ground coverage of one photo: L = l x S_num (l = photo side; S_num = 1/scale number)

Flying height H = f x S_num (above mean ground level)

Effective ground coverage (forward) = L (1 - Pf/100); (side) = L (1 - Ps/100)

Air base B = L (1 - Pf/100) ; Distance between flight lines W = L (1 - Ps/100)

Number of photos per strip = (strip length / B) + 1 (+ 2 extra at the ends)

Number of strips = (width of the area / W) + 1

Time interval between exposures = B / (speed of the aircraft)

Example

Area 12 km x 8 km; scale 1:15,000; photo 23 cm x 23 cm; f = 152 mm; Pf = 60%, Ps = 30%; aircraft speed 200 km/h.

L = 0.23 x 15,000 = 3450 m

H = 0.152 x 15,000 = 2280 m

B = 3450 x 0.4 = 1380 m ; W = 3450 x 0.7 = 2415 m

Photos per strip (12 km) = 12000/1380 + 1 = 8.7 + 1 = 10 (+2) = 12

Strips = 8000/2415 + 1 = 3.3 + 1 = 4.3 ~ 5

Time interval = 1380 / (200000/3600) = 24.8 sec

- Total photos = 12 x 5 = 60. Always allow extra photos for the ends of each strip, turning and wastage.

8. Stereoscopy

Stereoscopy is the science of seeing in three dimensions using two overlapping photographs of the same area taken from different positions (a **stereo pair**). Each eye sees one photograph, and the brain gives a 3-D image of the terrain (stereo model).

- **Stereoscopic vision:** the eyes see objects from slightly different directions; the **parallactic angle** gives the depth perception. Human eye base is 6.5 cm (about 65 mm).

Stereoscopes

Type	Description
Lens (pocket) stereoscope	Two simple lenses on a frame, about 5 cm apart, focal length 10 cm; small field; cheap and portable
Mirror stereoscope	Two large mirrors and two small ones; full overlap of 23 cm photos can be seen; with binoculars for magnification
Scanning / zoom stereoscopes	Higher magnification (zoom), used for the detailed interpretation
Anaglyph / polarised system	Red and blue (or polarised) images are separated by the coloured (polarised) glasses

Parallax and the height measurement

Stereoscopic (x-) parallax p of a point = the algebraic difference of the distances of its two images from the respective principal points, measured along the flight line (base line).

Absolute parallax of a point: $p = x_{left} - x_{right}$ (x in the photo coordinates measured on the flight line)

Elevation of a point: $h = H - (B f / p)$

Difference in elevation: $dh = (H - h_A)\, dp / (p_a + dp) = (H - h_A)\, dp / p_b$

- **Parallax bar (stereometer)** is used to measure the differences in parallax accurately (to 0.01 mm) while looking in the stereoscope: dp is the difference in the bar reading between two points.

- **Air base B** = ground distance between the two exposure stations; the photo base b = the average distance between the principal point and the conjugate principal point on the photographs: p = b for the point at the principal point.

> **NOTE:**
> Higher points have larger parallax, lower points have smaller parallax. Heights of objects from the parallax difference: h = (H x dp) / (p_base + dp) where p_base is the parallax at the base of the object.

Example

H = 2000 m above ground. Photo base b = 90 mm. The parallax difference between the top and the base of a tower = 1.2 mm. Height of tower:

$$h = H\,dp / (p_{base} + dp) = 2000 \times 1.2 / (90 + 1.2) = 26.3 \text{ m}$$

9. Ground Control Extension for Photographic Mapping

To plot photographs to a map scale we need ***ground control points (GCPs)*** - points identifiable on the photograph whose positions (X, Y, Z) are known from the ground survey (GPS, total station, levelling). Surveying every control point on the ground is costly. So control is ***extended*** from a few ground points by ***phototriangulation*** (aerial triangulation).

- ***Horizontal control:*** (X, Y) at least 2-3 points per model; ***vertical control:*** (Z) at least 3 points.
- ***Aerial triangulation*** gives the coordinates of the additional control points (***pass points / tie points***) in the strip or block from the measurements on the photos. It reduces the field work by a very large amount.

Types of aerial triangulation

Method	Principle
Radial triangulation (graphical / slotted templet)	Uses the property that angles at the principal point are true on a vertical photo (radial lines); intersection of rays gives horizontal positions
Analogue (instrumental) triangulation	Using stereo-plotting instruments, strips of the models are oriented successively

Analytical triangulation	Mathematical (least squares) adjustment of the strip / block with the photo coordinates; the ***bundle adjustment*** is the most accurate
GPS / INS supported triangulation	Exposure station coordinates and orientation from the onboard GPS and IMU reduce the ground control needs

Radial triangulation

Based on the principle that ***on a vertical photograph, the angles measured at the principal point (radial angles) are true horizontal angles*** (displacements due to tilt and relief are radial from the centre and do not change the angle).

- ***Graphical (paper templet) method:*** the principal points and the pass points are marked on each photograph; a transparent overlay (templet) is made with the radial lines from the principal point (centre) to each marked point. The templets of adjacent photographs are joined using the common points and the ground control, and the intersections give the unknown positions.
- ***Slotted templet method:*** the rays are represented by slots cut in a cardboard (or celluloid) templet; the studs fix the points. Assembled like a Meccano set to the ground control. Fast; used for large blocks.
- ***Mechanical (Arundel) method:*** like the slotted templet; ***Radial line plotter*** .
- Steps: (1) Select and mark the principal points, conjugate principal points, pass points and control points. (2) Draw radial lines. (3) Intersect the rays from the adjacent photos to get new points. (4) Scale and orient the strips to the ground control and adjust.

> ***NOTE:***
> Radial triangulation gives ***only plan (X, Y) positions*** ; elevations are got by the spirit levelling, or by the parallax method or the stereo-plotter.

10. Photographic Mapping

Maps can be produced from the aerial photographs by several methods, depending on the accuracy, cost, and the equipment available.

(a) Mapping using paper prints

- Photographic prints (contact prints at the photo scale) are used directly after ***rectification*** and the ***ratioing*** to the map scale; details are ***traced*** on the transparent overlay and the map is compiled with control.
- Planimetric map by the ***radial line method / sketchmaster / vertical sketchmaster / reflecting projector*** that enable the transfer of detail from the photo to the base map.
- Limitation: the displacements due to relief and tilt remain, so the accuracy is limited and the method suits flat terrain and small-scale maps.

(b) Mapping using stereo-plotting instruments

Stereo-plotters (stereo-plotting instruments) reconstruct the 3-D model from the overlapping photographs and let the operator trace the planimetric details and contours directly in the model with a floating mark. They give the ***accurate planimetric and contour maps*** .

Class	Examples / working
Direct optical projection	Multiplex, Balplex: diapositives projected optically; the floating mark on a platen is moved over the model
Mechanical projection	Wild A-series, Kelsh plotter: spatial rods represent the projected rays; model scale adjusted by the base
Analytical plotters	Computer-controlled; photo coordinates measured and mathematically transformed to the ground coordinates (high precision)
Digital photogrammetric workstations (softcopy)	Digital images on a computer; automated matching to generate the DEM, orthophotos, contours, 3-D models

- Stages: ***interior orientation*** (restoring the camera geometry), ***relative orientation*** (removing the y-parallax to form the model) and ***absolute orientation*** (scaling and levelling the model using the ground control).
- Floating mark: the operator moves a mark in x, y, z so that it appears to rest on the model; the instrument records / plots the position -> planimetry and contours.

(c) Mosaics

A ***mosaic*** is an assembly of overlapping aerial photographs (trimmed and joined) to form a continuous picture of the area.

- ***Controlled mosaic:*** made from rectified and ratioed photos laid to the ground control; accurate; scale is uniform.
- ***Uncontrolled mosaic:*** made by matching the details of the adjacent photos, without any control; quick and cheap; not accurate.
- ***Semi-controlled mosaic:*** a combination – the control is used for some portions.
- Methods: the central portion of each photo is used (to reduce the displacements); cut along the patterns / feathered edges; photographed to the final scale.
- Uses: quick view / planning, interpretation of the whole area, as a map substitute, base for the thematic mapping.

(d) Map substitutes

Products made from the aerial photographs that can be used ***instead of a line map*** , when quick and economical information is needed.

- ***Photomap:*** a mosaic or photograph with the grid, names and the symbols added – map substitute for rapid planning.
- ***Orthophotograph / orthophoto map:*** photograph in which the relief and tilt displacements have been removed (differential rectification using a DEM) – so it has a uniform scale like a map; with contours it is an orthophoto map.
- ***Mosaic (controlled / uncontrolled) maps*** and ***stereo-pairs*** (for the interpretation).

> **KEY:**
> Orthophoto = photograph + map. It has the details of a photo and the accuracy of a map (uniform scale). Digital orthophotos are produced by the softcopy workstations / drone processing.

II. Photo Interpretation (brief)

- Identification of objects from their ***tone, texture, shape, size, shadow, pattern, location and association*** .
- Uses in geology, forestry, agriculture, hydrology, urban and transport planning, and disaster assessment.

12. Modern Trends

- Digital cameras, ***drones*** (UAV photogrammetry), ***GNSS-INS*** (direct georeferencing), ***LiDAR*** , satellite stereo (Cartosat, SPOT, WorldView) for DEM and mapping; automatic image matching and ***Structure-from-Motion*** .
- Software products: Pix4D, Agisoft Metashape, ERDAS, Trimble Inpho – produce orthomosaics, DSM and 3-D models in a short time.

13. Important Questions to Prepare

- Define photogrammetry. Explain the classification of photographs and the terms on a vertical photograph.
- Derive the expression for the scale of a vertical photograph and the relief displacement.
- Explain the tilt displacement and the effect of the tilt on a photograph.
- What are the factors in the flight planning? Calculate the number of photos for a given area.
- Describe the stereoscope types, the parallax bar and the use for finding the heights.
- Explain the radial triangulation methods. Write notes on mosaics and map substitutes.

14. Quick Revision

- Scale = f/(H – h). Relief displacement d = r h / H. Overlaps: 60 % forward, 30 % side.
- Height from the parallax: dh = (H – h) dp / (p + dp). Air base B = L (1 – Pf/100).
- Radial triangulation uses angles at the principal point which are true on a vertical photo.
- Stereo-plotter orientations: interior, relative, absolute. Orthophoto removes the relief and tilt displacements.